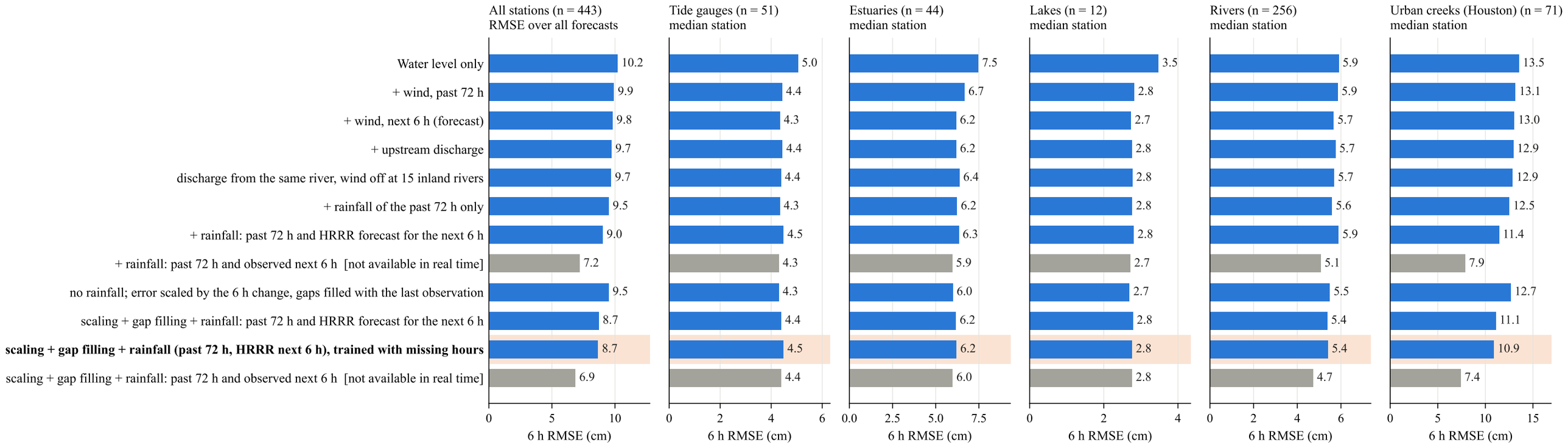
All stations (n = 443)
RMSE over all forecasts
Tide gauges (n = 51)
median station
Estuaries (n = 44)
median station
Lakes (n = 12)
median station
Rivers (n = 256)
median station
Urban creeks (Houston) (n = 71)
median station
Water level only
+ wind, past 72 h
+ wind, next 6 h (forecast)
+ upstream discharge
discharge from the same river, wind off at 15 inland rivers
+ rainfall of the past 72 h only
+ rainfall: past 72 h and HRRR forecast for the next 6 h
+ rainfall: past 72 h and observed next 6 h [not available in real time]
no rainfall; error scaled by the 6 h change, gaps filled with the last observation
scaling + gap filling + rainfall: past 72 h and HRRR forecast for the next 6 h
scaling + gap filling + rainfall (past 72 h, HRRR next 6 h), trained with missing hours
scaling + gap filling + rainfall: past 72 h and observed next 6 h [not available in real time]
10.2
9.9
9.8
9.7
9.7
9.5
9.0
7.2
9.5
8.7
8.7
6.9
5.0
4.4
4.3
4.4
4.4
4.3
4.5
4.3
4.3
4.4
4.5
4.4
7.5
6.7
6.2
6.2
6.4
6.2
6.3
5.9
6.0
6.2
6.2
6.0
3.5
2.8
2.7
2.8
2.8
2.8
2.8
2.7
2.7
2.8
2.8
2.8
5.9
5.9
5.7
5.7
5.7
5.6
5.9
5.1
5.5
5.4
5.4
4.7
13.5
13.1
13.0
12.9
12.9
12.5
11.4
7.9
12.7
11.1
10.9
7.4
0
5
10
0
2
4
6
0.0
2.5
5.0
7.5
0
2
4
0
2
4
6
0
5
10
15
6 h RMSE (cm)
6 h RMSE (cm)
6 h RMSE (cm)
6 h RMSE (cm)
6 h RMSE (cm)
6 h RMSE (cm)